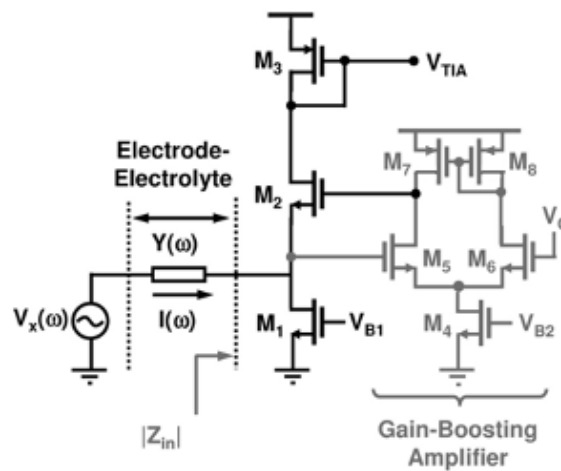


Design of a size-aware in-pixel current conveyor (TIA)

Deadline 09/11/2025

For this assignment, you will design the current conveyor stage assisted by a gain-boosting amplifier in the figure below. You will need to size all transistors and simulate the circuit. Finally, you will need to write a report where you describe your design choices and the performance you achieved. **Check the lecture slides for details on the report.** You can use the tool of preference for the report (e.g. LaTeX, Word, Gdoc), but the final report has to be a PDF file. For the circuit simulation, you will use the same technology that you have been using throughout the class. **You need to submit your simulation schematics as well.** Failure to do so will result in a lower grade, even if the report is perfect.



The following specifications are required:

- $G_{TIA} > 10 \text{ k}\Omega$
- $BW > 100 \text{ kHz}$
- $Z_{IN} < 1 \text{ k}\Omega$
- Noise $I_{in,n} < 1 \text{ nA}_{rms}$
- Estimated pixel area (excluding sensor) $< (50 \text{ }\mu\text{m} \times 50 \text{ }\mu\text{m})$
- $VDD=1.8\text{V}$
- $C_L=1\text{pF}$ (at the output of the TIA)

Here are a few hints for the assignment:

- Replace the voltage input source and electrode-electrolyte impedance (V_x and $Y(\omega)$) with a current source in parallel with $10 \text{ M}\Omega$ and 0.5 pF load for simulation
- Start without the amplifier and bias M_2 with fixed potential
- The gain is in Ω as it's a current conveyor with a load (input is current, output is voltage)
- Remember to consider the input impedance of the diode-connected MOS M_3
- Reserve sufficient time for the report writing and make sure to present your results in a clear and organized manner